Reverse-Engineering Mathematical Reasoning from OpenAI's 2026 Mathematics Corpus: Implications for Mechanistic Interpretability

Taechasith Kangkhuntod
School of Engineering, University of the Thai Chamber of Commerce, Thailand

October 2026

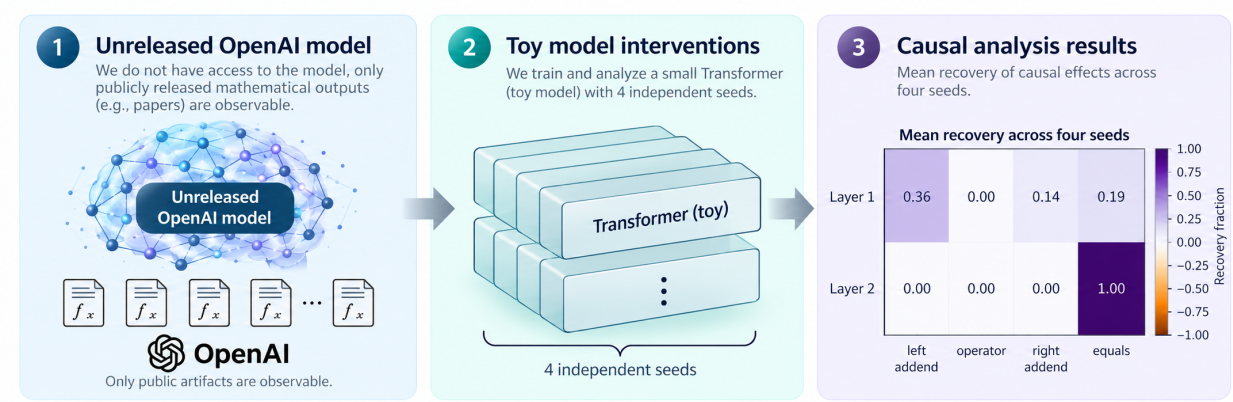


Figure 1: Overall conceptual workflow. The public OpenAI mathematics release supports artifact reconstruction and bounded held-out prediction, but does not expose model internals. Controlled hidden-process model organisms calibrate what output-only reconstruction can identify. Causal interventions, if an eligible inspectable model is available, remain a separate evidence stream.

Abstract

OpenAI's public 2026 mathematics release is an unusually rich public record: mathematical manuscripts, source artifacts, Lean materials, and ten abridged reasoning summaries. It is not a release of model weights, activations, full prompts, or complete internal reasoning. We present a reproducible, preregistered artifact audit at a pinned release commit. We find 372 result families and 719 catalogue-linked manuscripts, with every linked PDF and TeX-source directory present in the Git tree. We identify 242 families with a Lean scope document and 10 released abridged summaries. Catalogue-linked manuscript count does not meaningfully differ by scope-document availability (Mann–Whitney $p = 0.638$; permutation $p = 0.665$; rank-biserial $r = 0.03$). In a separately specified exploratory extension, we structurally index every versioned Git blob (133,955 blobs; 2.06 GiB) by path, object identifier, size, and artifact role. In a Version 2 exploratory continuation, we further extract 45,620 literal TeX environments and 97,343 within-document source references from 8,007 source files, while separately retaining 42,143 rule-based motif candidates. These structures are not proof verification or semantic annotation. A frozen, family-disjoint prediction experiment finds that literal-reference graph features predict the lexical *induction* label with AUROC 0.74, versus 0.71 for matched metadata; this is an association with an extraction label, not a recovered mathematical strategy. Controlled model organisms further show that distinct hidden producers with identical observable laws

cannot be identified (0.46 accuracy against 0.50 chance). We release code, provenance hashes, derived data, figures, and a venue-neutral reproducibility bundle. No result identifies OpenAI's hidden reasoning or neural mechanisms.

1 Introduction

Reasoning traces expose intermediate claims, proof organization, and references to verification. Yet a persuasive chain of thought is not automatically a faithful account of the internal computation that produced an answer [14, 30]. Mechanistic interpretability requires internal access and causal intervention, as in work on sparse features and circuit-level representations [8, 29].

OpenAI's October 2026 public mathematics repository contains manuscripts and supporting artifacts produced by an unreleased internal model [26, 27]. It is valuable for studying a public record of model-assisted mathematics, but it should not be read as a transparent window into neural computation. We provide a pinned inventory, a preregistered artifact-level comparison, and descriptive measurements of the published summaries. We then separately demonstrate a genuine causal mechanistic-interpretability intervention in a toy model.

2 Scientific context: mechanistic interpretability

Mechanistic interpretability seeks explanations expressed in a model's actual computations: features, states, edges, and algorithms that can be interrogated and causally manipulated. The circuit perspective developed from feature visualization and transformer-circuit analyses [7, 23, 24] is deliberately stronger than a useful behavioral description. It asks whether a proposed mechanism predicts the effects of intervention.

There are instructive examples in accessible models. Work on induction heads, modular-addition grokking, indirect-object identification, and a GPT-2 greater-than capability combines structure, activations, and interventions to propose circuit-level accounts [13, 22, 25, 32]. Those successes are neither a license to infer circuits from prose nor a guarantee that a localized explanation is complete. They instead specify the kind of evidence missing for an unreleased model.

Causal tools make the distinction concrete. Causal mediation, activation and path patching, model editing, automated circuit discovery, and causal scrubbing each test particular counterfactual claims [4, 5, 10, 11, 20, 31, 34]. They also expose failure modes: an ablated computation can be compensated by others, and a single intervention can overstate a component's explanatory role [2, 9, 19].

Recent benchmarks sharpen the evaluation standard. KisMATH aligns extracted reasoning graphs with interventions on mathematical tasks, while MIB compares mechanistic localization methods against shared causal targets [21, 28]. These works rule out a priority claim for graph-informed mathematical interventions here. Instead, the Version 2 continuation asks whether public research-source structure can generate family-held-out causal hypotheses in a separately specified inspectable model.

Feature-level approaches address the fact that individual neurons can mix several concepts. Toy superposition models, dictionary learning, and sparse autoencoders supply hypotheses about more useful feature bases [3, 6, 8, 29]. In parallel, chain-of-thought and process-supervision research clarifies why visible reasoning can aid evaluation while remaining distinct from the latent computation that determined an answer [14–16, 30, 33]. Our simulation adopts the causal-intervention side of this program while making no claim to have recovered an arithmetic circuit.

3 Evidence boundary

4 Identifiability and evidential boundaries

Our framework separates artifact reconstruction (Level A), validated strategy reconstruction (Level B), held-out functional prediction (Level C), causal validation in an inspectable model (Level D), and target-producer mechanism identification (Level E). Only Level A is fully available from this release. Version 3 tests a deliberately weaker Level-C proxy: whether literal source-graph features predict an already-fixed lexical extraction label in family-disjoint public documents. It cannot validate a proof strategy because the labels have not received independent semantic adjudication.

This separation follows causal-abstraction theory and benchmark work requiring intervention against known causal targets [10, 21]. KisMATH demonstrates graph-aligned interventions in accessible models, but uses reasoning graphs and open models rather than a selected publication archive [28]. Public mathematical prose is not a faithful readout of hidden computation merely because it is coherent [30]. Consequently, no arrow in Figure 1 is an arrow into the unreleased OpenAI producer.

The release contains public outputs and supporting artifacts. It does not release weights, activations, attention patterns, optimizer state, complete prompts, training data, or complete hidden reasoning. Therefore, our strongest permissible claims concern availability, organization, and surface-text statistics. We cannot identify a circuit, feature, attention head, latent representation, or causal sequence in the OpenAI model. Nor can a manuscript or a formalization scope page establish the correctness of a mathematical claim. This distinction matters because process supervision can make intermediate steps useful targets without making prose a complete causal explanation [15].

The title's implication for mechanistic interpretability is methodological. The public record can motivate frozen benchmarks and hypotheses for inspectable models, such as whether a proof segment is causally necessary for a conclusion. It cannot answer that question without a model whose internal states can be recorded and manipulated.

5 Corpus and method

5.1 Frozen source

We acquired the Apache-2.0 licensed `openai/math` repository at commit `fd4aeeb2...0529eeeb`; the acquisition manifest records the full 40-character identifier. The manifest records SHA-256 hashes of `README.md`, `CONTENTS.md`, and `LICENSE`, plus a successful `git fsck`. Windows path limits prevent materialization of a few extremely deep upstream paths in a normal working tree. Rather than silently lose them, the pipeline reads immutable Git blobs directly from the verified object database.

The pre-analysis contract is in `RESEARCH_PROTOCOL.md` and `PREREGISTRATION.md`. The unit is a result family for catalogue measures and a manuscript for file availability. Families are not independent model runs.

5.2 Measurements

We parse each numbered family and each linked PDF in `CONTENTS.md`. For each manuscript, we measure presence of the linked PDF, a non-vendor TeX file, and a README marker. For each family, we measure whether `lean/docs/NNN.md` exists. This is Lean scope-document availability,

not verified correctness or successful compilation. For the ten summary PDFs named in the upstream README, we extract pages, word-like tokens, and transparent keyword counts. Failures remain in a warnings table.

The sole preregistered comparison asks whether catalogue-linked manuscript count differs between families with and without a Lean scope document. We report medians, a two-sided Mann–Whitney test, rank-biserial correlation, and a two-sided seed-20261009 label-permutation calculation using 10,000 draws. The latter is a reference calculation, not a claim of independent random families.

5.3 Exhaustive release-tree extension

The catalogue measures do not by themselves cover every versioned artifact. We therefore registered a separate exploratory protocol before executing a structural census of every Git blob at the same immutable commit (`FULL_TREE_INVENTORY_PROTOCOL.md`). The implementation resolves the pinned commit once, reads immutable Git objects rather than the working tree, and records each blob's path, object identifier, byte size, top-level directory, extension, deterministic path/extension-based artifact role, and catalogue or released-summary relation. This is exhaustive at the release-tree level: it includes uncatalogued preprint bundles, sources, configuration, and release metadata rather than treating the catalogue as the denominator. It is not a semantic reading of every file, a proof check, Lean compilation, or an expansion to all research artifacts OpenAI may publish elsewhere.

5.4 Version 2 exploratory source-structure extraction

Before any Version 2 extraction, we froze a continuation protocol and a separate inactive confirmatory plan. The extractor reads immutable Git blobs for all preprint TeX sources. It records TeX theorem-, lemma-, proposition-, corollary-, definition-, remark-, and proof-environment candidates with source commit, blob identifier, document identifier, result-family relation, and line span. Graph A retains only literal within-document TeX references. Graph B contains fixed lexical rules for candidates such as contradiction, induction, construction, reduction, invariant, case splitting, limiting argument, and equivalence. Graph A is not an entailment graph; Graph B is not semantic or human annotation.

This checkpoint deliberately does not create validated mathematical counterfactuals or execute a new model intervention. Lean kernel verification does not by itself certify informal semantic fidelity [1]; without a detected Lean toolchain and independent semantic audit records, no Version 2 counterfactual is labelled verified.

6 Results

6.1 Release inventory

Artifact	Count
Result families	372
Catalogue-linked manuscripts	719
Families with Lean document	242
Linked manuscript PDFs present	719
Manuscripts with TeX source	719
Released abridged summaries	10

The parser recovered 372 families and 719 linked manuscripts, matching the upstream catalogue headline. Every linked PDF and manuscript directory with TeX sources was present in the pinned Git tree (Figure 3). There are 242 families with a Lean scope document (65.1%). OpenAI separately reports approximately 42% of top-line results as formalized [27]; this uses a different unit and is not interchangeable with our scope-document measure.

6.2 Exhaustive release-tree census

The same release tree contains 133,955 versioned blobs totaling 2.06 GiB. The structural census includes 122,876 Lean source files, 8,007 preprint TeX sources, and 776 preprint PDFs (Table 1; Figure 2). These are file counts and path-based roles, not claims that the Lean files compiled, the manuscripts are correct, or each source file was semantically interpreted.

Exhaustive tree inventory	Count
All Git blobs	133,955
Lean source files	122,876
Preprint TeX source files	8,007
Preprint PDFs	776
Named released summaries	10

Table 1: Exhaustive structural inventory of every versioned blob in the pinned release. Categories are deterministic path/extension roles and their units are files, not mathematical results.

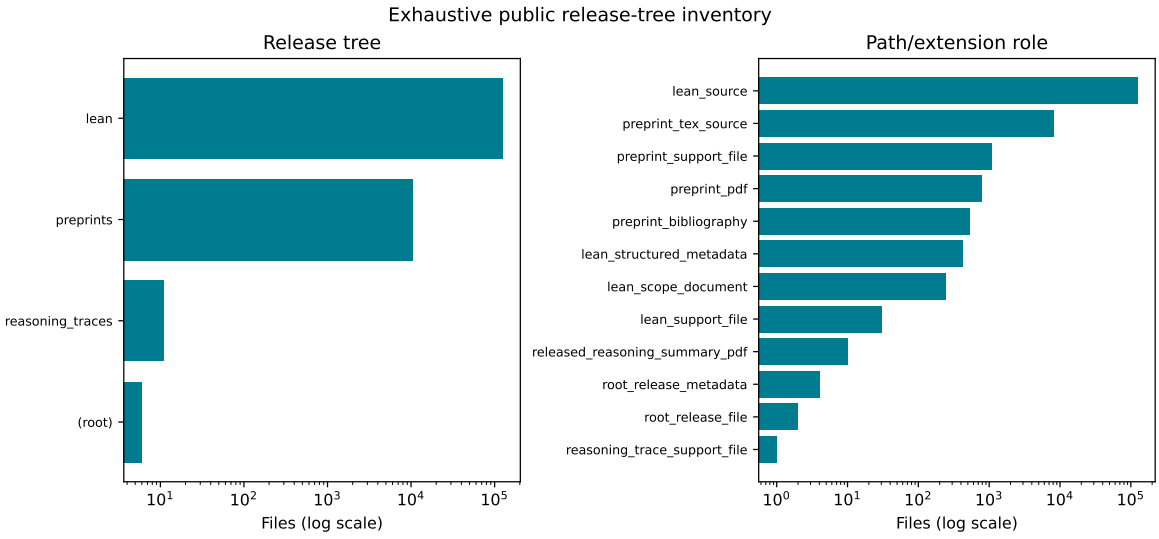


Figure 2: Complete path/extension-based inventory of the pinned release tree. The log scale makes the small root and reasoning-summary groups visible beside the large Lean and preprint collections. This is a structural census of all versioned blobs, not a quality or correctness assessment.

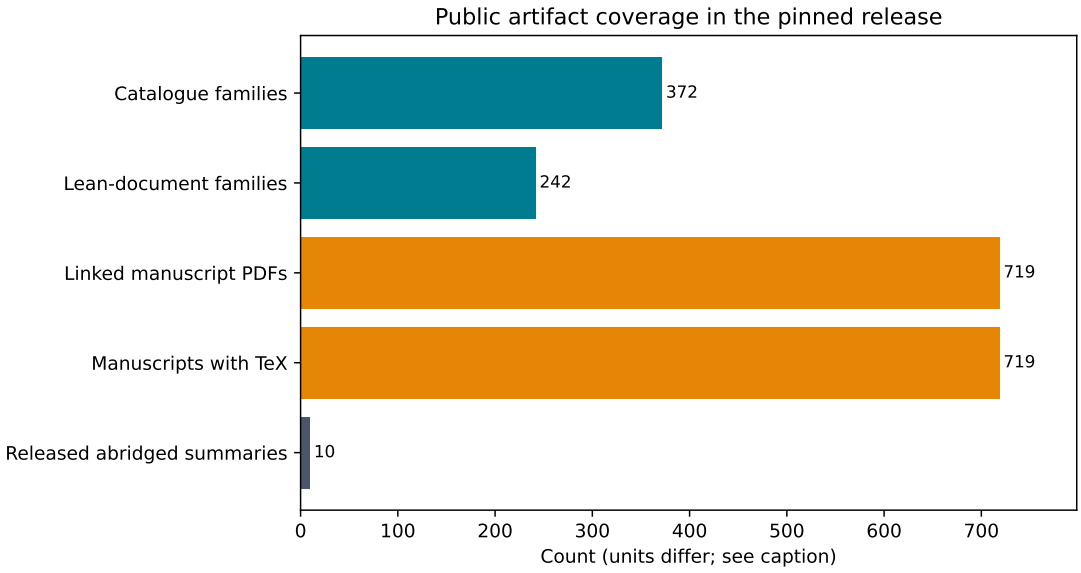


Figure 3: Public artifact coverage in the pinned release. Bars mix family, manuscript, and summary units, so they describe availability rather than a common rate or quality score.

Family size is concentrated at one linked manuscript, with a smaller tail of families containing companion arguments or consequences (Figure 4). This is a catalogue-structure observation, not a count of independent solution attempts.

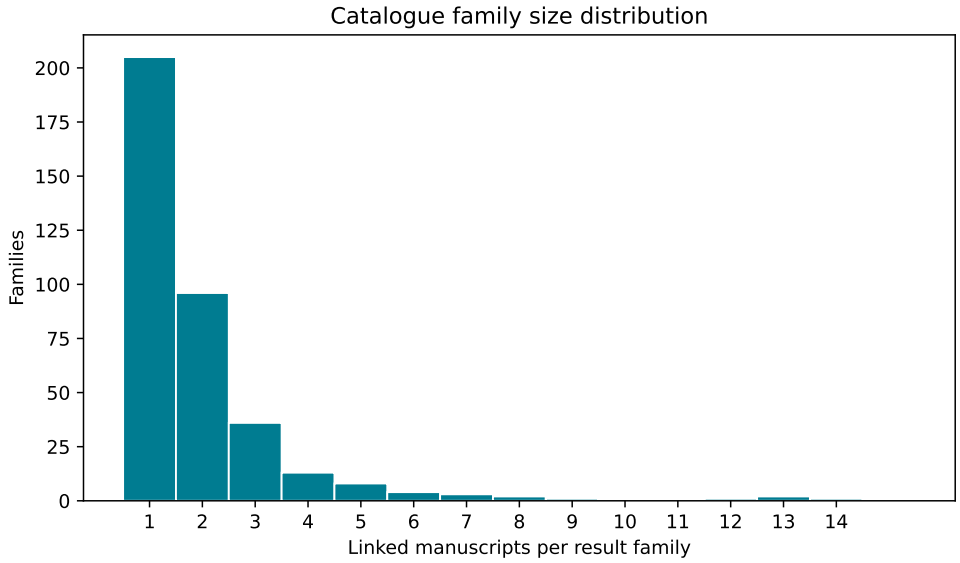


Figure 4: Distribution of catalogue-linked manuscripts per result family.

6.3 Scope documents and family size

Both groups have a median of one linked manuscript. The comparison is small and non-significant ($U = 16{,}150$, $p = 0.638$, $r = 0.03$); the deterministic permutation calculation gives $p = 0.665$. This does not demonstrate equality, causal independence, mathematical quality, or Lean compilation

success. It only gives no meaningful evidence that this simple catalogue-size attribute separates families by scope-document availability at this release.

6.4 Released summaries

The 10 named summaries span 5–45 PDF pages and 2,020–19,935 extracted word-like tokens (Figure 8). Their non-random selection and heterogeneity make them unsuitable for estimating properties of all model reasoning. They can support qualitative study of the published text but cannot recover omitted deliberation or establish faithfulness to unobserved activations.

6.5 Exploratory proof-structure candidates

The Version 2 extractor processed 8,007 TeX files, yielding 45,620 source-environment candidates, 97,343 deduplicated literal reference edges, and 42,143 lexical motif records. It could not resolve 15,824 labels within their source document; those references are excluded rather than guessed. Figure 5 is a frequency display of rule matches, not a prevalence estimate for proof strategies. Repeated document templates, prose conventions, and unequal source lengths may all influence these counts.

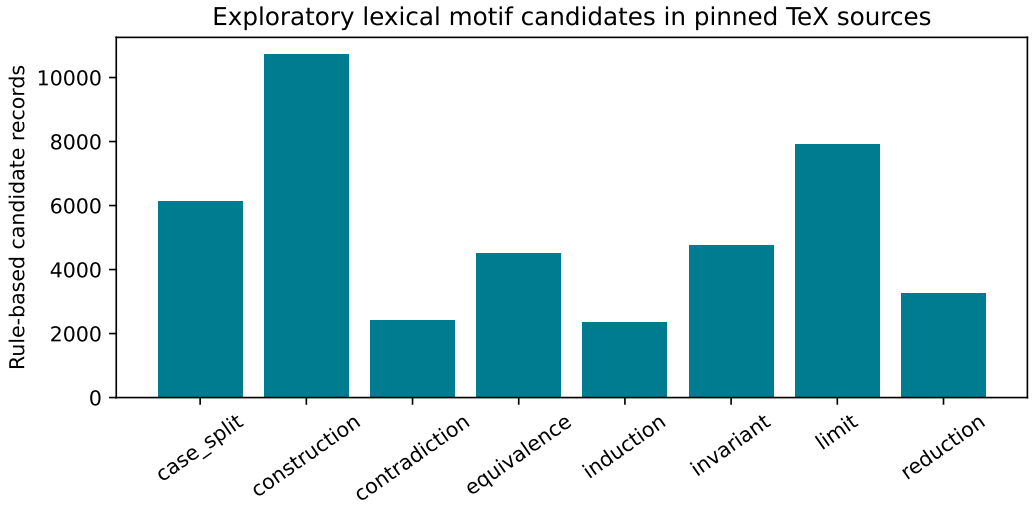


Figure 5: Exploratory lexical proof-motif candidates from pinned TeX sources. Each bar counts a fixed regular-expression match in a candidate environment; it is neither a semantic annotation nor evidence of a mathematical dependency.

6.6 Family-disjoint functional reconstruction proxy

We froze a single exploratory prediction task before inspecting Version 3 outcomes. The target is the extractor's fixed lexical *induction* label, not a semantic strategy label. The unit is an extracted node; the held-out unit is a result family, with each unlinked document kept as its own group. The structural logistic model receives node kind, within-document position, document size, and literal-reference in/out degree. It never receives source text, the matching words, document identifiers, or the target label. Matched metadata-only and degree-permuted-structure controls use the same split and model capacity.

The held-out set contains 8,541 nodes from 80 groups, with a 5.5% induction label rate. The structural model obtains AUROC 0.74, versus 0.71 for metadata and 0.71 after degree permutation. The structure-minus-metadata AUROC difference is 0.028 (group bootstrap 95% CI 0.010–0.045). Figure 6 visualizes the fixed comparison. This modest association survives these controls, but does not separate reusable proof strategies from mathematical conventions, editing, or source-selection effects. It is therefore not evidence for Level-B strategy reconstruction.

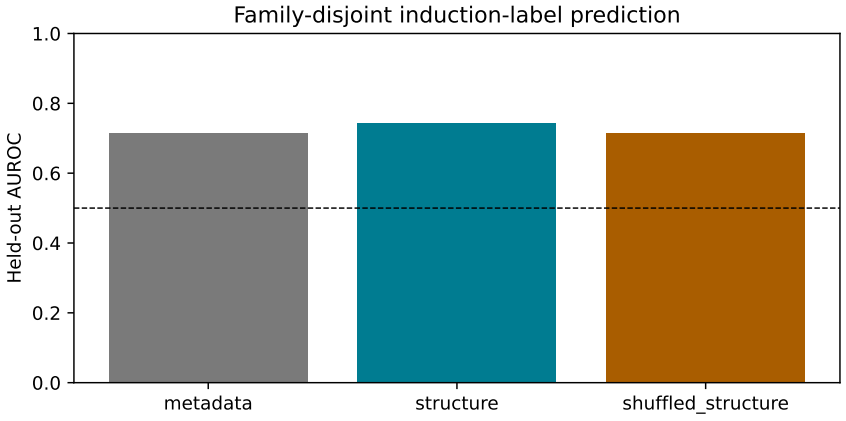


Figure 6: Held-out prediction of a lexical *induction* extraction label. The graph-feature model exceeds matched metadata and degree-permuted controls, but the target remains a reproducible source-text label rather than a verified proof strategy.

6.7 Controlled hidden-process identifiability

To test whether an output-only reconstruction can identify an otherwise hidden producer, we generated 1,440 synthetic artifacts from six known process families, froze their hidden labels, and supplied the reconstruction classifier only observable motif, reference-count, length, and node-kind fields. Overall held-out process accuracy was 0.72. Crucially, two different producers were defined with exactly the same observable law. Their identity accuracy was 0.46, below the 0.50 chance reference (Figure 7). The controlled negative result establishes an equivalence class: distinct processes cannot be recovered when the available observations are deliberately identical. It calibrates an identifiability limitation; it says nothing about which process produced the OpenAI corpus.

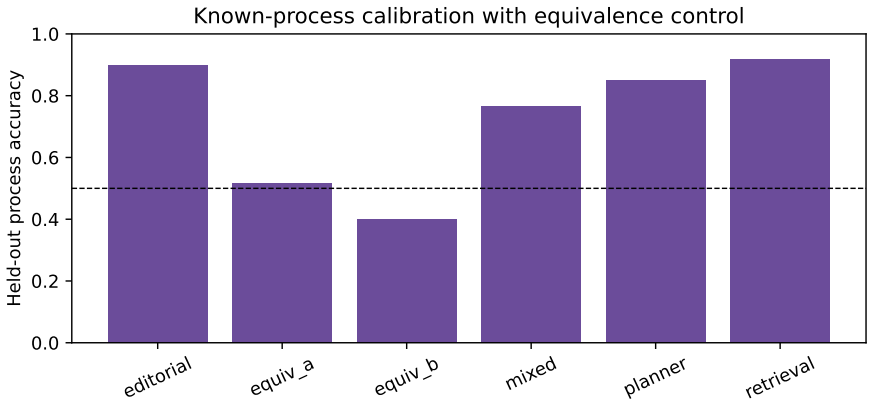


Figure 7: Known-process model organisms. Easily distinguishable synthetic producers can be classified, while the intentionally observationally equivalent pair cannot. This is a calibration experiment, not an attribution test for OpenAI.

6.8 Finite arithmetic counterfactual checks

We also generated 480 concrete integer-arithmetic variants of 120 base examples: original, commuted, changed-hypothesis, and invalid-conclusion forms. Exact Python integer checks agreed with all declared statuses (360 valid and 120 invalid). This is a finite computational check, not a Lean-kernel certificate or an audit of informal semantic fidelity. It is therefore excluded from causal evaluation, consistent with documented formal benchmark failure modes [1].

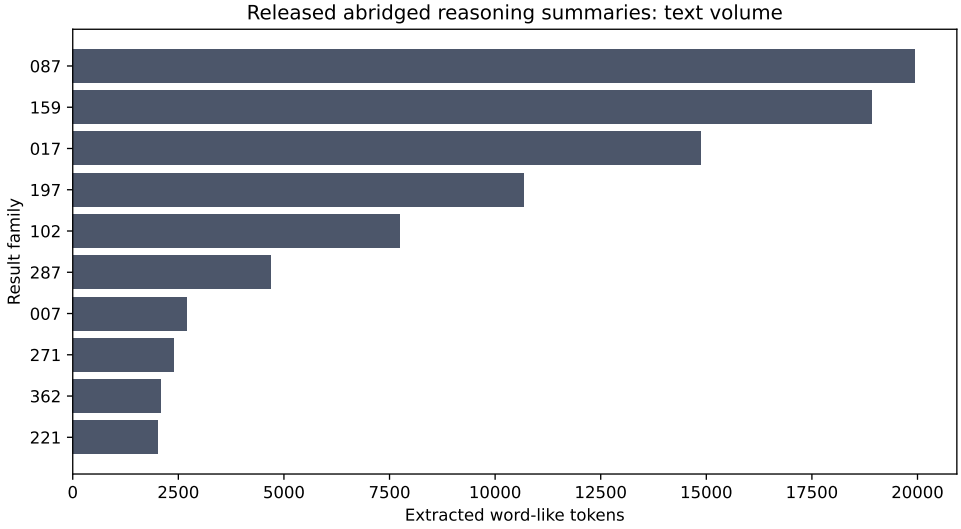


Figure 8: Extracted word-like token volume of the ten explicitly released abridged summaries. This is descriptive parser output, not a reasoning-quality or faithfulness metric.

7 Causal intervention in an inspectable surrogate

To distinguish output analysis from a mechanistic experiment, we preregistered the extension in `SIMULATION_PROTOCOL.md` before running 4 independently initialized 39,264-parameter, two-layer,

three-head Transformers from scratch on all 100 ordered pairs of addition modulo 10. Their frozen vocabulary has digits, a plus token, and an equals token; every seed reached 100% training accuracy. This deliberately small, in-sample task is not a benchmark of mathematical reasoning or generalization.

For each clean prompt $(a, +, b, =)$, we corrupted the left addend with $a + 1 \pmod{10}$. At each layer and position, we replaced the corrupted post-block residual vector with the clean vector and measured recovery of the clean answer logit relative to clean and corrupted baselines. This is an activation-patching intervention [10, 34]. For prompt i, recovery is

$$R_i = \frac{\ell_y(\text{patched}, i) - \ell_y(\text{corrupt}, i)}{\ell_y(\text{clean}, i) - \ell_y(\text{corrupt}, i)}.$$

Near-zero denominators are excluded under the fixed protocol. We average R_i within model and use the four initialization seeds as the uncertainty unit, reporting two-sided t intervals with three degrees of freedom.

Layer 1 left-addend patching recovered 0.36 of the clean-logit gap (95% CI [0.20, 0.52]), compared with 0.00 (95% CI [0.00, 0.00]) for the contextual operator control. The preregistered paired difference is 0.36 (95% CI [0.20, 0.52]). Layer 2 equals-position patching recovered 1.00 (95% CI [1.00, 1.00]; Figure 9). The last result is an outcome-position positive control because the unembedding reads that final residual position; it is not evidence for a distributed arithmetic circuit.

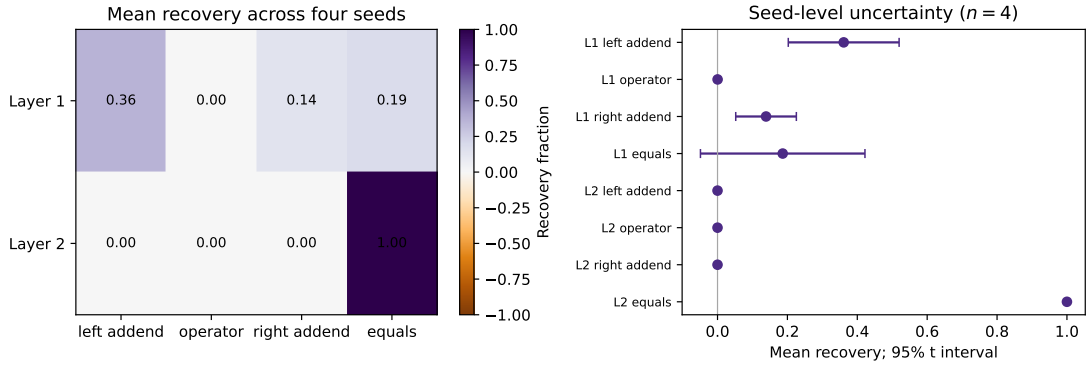


Figure 9: Four-seed causal residual-stream patching in the inspectable toy Transformer. The heat map gives mean clean-answer-logit recovery after inserting a clean residual state into a left-addend-corrupted run; points show the mean and two-sided 95% Student-t interval across four independent initializations. This figure is separate from the OpenAI artifact audit.

The surrogate result demonstrates a repeatable causal-intervention procedure. It does not establish a reusable arithmetic circuit, necessity or sufficiency of any site, generalization beyond its training table, or any property of OpenAI's model.

8 Limitations and research agenda

This audit concerns one release commit; catalogue labels are upstream metadata, not independent adjudication. The exhaustive tree census covers every versioned blob structurally, but it does not read every file semantically. We did not compile the substantial upstream Lean project or assess mathematical validity. PDF extraction is imperfect, although failures are recorded. The toy Transformer was trained on its evaluation table, so its result is methodological rather than evidence

of abstract reasoning. Most importantly, public output artifacts cannot identify mechanisms inside an unreleased model.

Version 3 remains limited. Its source-structure outputs meet corpus integrity and provide a documented exploratory input for G2, but they have not been independently semantically audited. The finite arithmetic cases give only a partial G3 check; they are not Lean-checked. The family-disjoint label prediction and model-organism equivalence control make G7 partial. G4–G6 are blocked: no eligible open-weight model, behavioral-competence result, or new causal intervention exists. CPU-only PyTorch, no cached open-weight model, and no detected Lean toolchain are recorded infrastructure constraints, not negative scientific findings.

A credible next study should choose an open-weight mathematical model, freeze prompts and scoring, collect clean and controlled-corruption runs, and test causal effects through activation patching, path patching, ablation, or feature intervention across seeds and held-out tasks. It should explicitly separate a visible rationale from both the answer and the latent state, and test proposed circuits against multi-site or resampling interventions rather than treating a single patching map as confirmation.

A useful shared next artifact would be a versioned causal-transfer benchmark: a corpus-derived task taxonomy, fixed prompt and counterfactual generator, formal or otherwise transparent scoring suite, model-and-seed matrix, and intervention registry all frozen before outcomes are observed. It should retain only candidate explanations that predict intervention effects across model families and held-out task variants, and document licensing and training-data contamination risks. Such a benchmark would enable task-specific comparisons of sparse probes, sparse feature circuits, and feature dictionaries under shared control and interpretability criteria [12, 17, 18]. This is a proposed research infrastructure, not a claim that the current audit has identified an OpenAI circuit.

9 Conclusion

The pinned 2026 mathematics release is not a window into an unreleased model's circuitry. It is, however, a durable public specification of a demanding mathematical-output domain: 372 result families, 719 linked manuscripts, Lean scope materials, and 10 released summaries, now situated within an exhaustive 133,955-blob release-tree census. This study releases a pinned, auditable map of that domain together with a deliberately separate causal-intervention template.

That division makes the work actionable. Future studies can preregister behavioral and structural hypotheses from the public record; instantiate matched prompts, perturbations, and scoring rules in inspectable open-weight mathematical models; and retain only candidate explanations that predict intervention effects across seeds, model families, and held-out task variants. Such studies would not identify OpenAI's mechanism. They could establish the independently valuable result that particular causal computational strategies are sufficient and robust within the tested model population for behaviors the release makes observable.

The contribution is therefore a reproducible bridge from public mathematical artifacts to falsifiable research on source structure. The bounded predictive gain is retained as an empirical result, while the controlled equivalence failure prevents it from becoming a claim about the producer process. Interventions in accessible models would still be needed for a mechanistic claim. Keeping these evidence types separate makes the work cumulative rather than promotional.

Reproducibility and ethics

Derived outputs are regenerated by the analysis, surrogate, and Version 2 proof-extraction modules documented in the repository README. The source archive excludes third-party corpus files. The full-tree census is specified in the accompanying inventory protocol. This study makes no claim about proprietary model internals, intent, or hidden reasoning. The complete research repository is available at the project repository URL.

References

[1] Pawan Sasanka Ammanamanchi, Siddharth Bhat, and Stella Biderman. Faults in our formal benchmarking: Dataset defects and evaluation failures in lean theorem proving. *arXiv preprint arXiv:2606.29493*, 2026. URL https://arxiv.org/abs/2606.29493.

[2] Anthony Bau, Yonatan Belinkov, Hassan Sajjad, Fahim Dalvi, Nadir Durrani, and James Glass. Identifying and controlling important neurons in neural machine translation. In *International Conference on Learning Representations*, 2019. URL https://arxiv.org/abs/1811.01157.

[3] Trenton Bricken, Adly Templeton, Joshua Batson, et al. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. URL https://transformer-circuits.pub/2023/monosemantic-features/index.html.

[4] Lawrence Chan, Adrià Garriga-Alonso, Nicholas Goldowsky-Dill, et al. Causal scrubbing: a method for rigorously testing interpretability hypotheses. AI Alignment Forum, 2022.

[5] Arthur Conmy, Augustine N. Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. In *Advances in Neural Information Processing Systems*, 2023. URL https://arxiv.org/abs/2304.14997.

[6] Hoagy Cunningham et al. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023.

[7] Nelson Elhage, Neel Nanda, Catherine Olsson, et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. URL https://transformer-circuits.pub/2021/framework/index.html.

[8] Nelson Elhage, Tristan Hume, Catherine Olsson, et al. Toy models of superposition. *Transformer Circuits Thread*, 2022. URL https://transformer-circuits.pub/2022/toy_model/index.html.

[9] Matthew Finlayson, Aaron Mueller, Sebastian Gehrmann, Stuart Shieber, Tal Linzen, and Yonatan Belinkov. Causal analysis of syntactic agreement mechanisms in neural language models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, 2021. URL https://aclanthology.org/2021.acl-long.144/.

[10] Atticus Geiger, Duligur Ibeling, Amir Zur, et al. Causal abstraction: A theoretical foundation for mechanistic interpretability. *Journal of Machine Learning Research*, 26(83):1–64, 2025. URL https://jmlr.org/papers/v26/23-0058.html.

[11] Nicholas Goldowsky-Dill, Chris MacLeod, Lucas Sato, and Aryaman Arora. Localizing model behavior with path patching. *arXiv preprint arXiv:2304.05969*, 2023. URL https://arxiv.org/abs/2304.05969.

[12] Wes Gurnee, Neel Nanda, Matthew Pauly, Katherine Harvey, Dmitrii Troitskii, and Dimitris Bertsimas. Finding neurons in a haystack: Case studies with sparse probing. *arXiv preprint arXiv:2305.01610*, 2023. URL https://arxiv.org/abs/2305.01610.

[13] Michael Hanna, Ollie Liu, and Alexandre Variengien. How does GPT-2 compute greater-than?: Interpreting mathematical abilities in a pre-trained language model. *arXiv preprint arXiv:2305.00586*, 2023. URL https://arxiv.org/abs/2305.00586.

[14] Tamera Lanham, Anna Chen, Ansh Radhakrishnan, et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*, 2023. URL https://arxiv.org/abs/2307.13702.

[15] Hunter Lightman, Vineet Kosaraju, Yura Burda, et al. Let's verify step by step. *arXiv preprint arXiv:2305.20050*, 2023. URL https://arxiv.org/abs/2305.20050.

[16] Qing Lyu, Shreya Havaldar, Adam Stein, et al. Faithful chain-of-thought reasoning. *arXiv preprint arXiv:2301.13379*, 2023. URL https://arxiv.org/abs/2301.13379.

[17] Aleksandar Makelov, George Lange, and Neel Nanda. Towards principled evaluations of sparse autoencoders for interpretability and control. *arXiv preprint arXiv:2405.08366*, 2024. URL https://arxiv.org/abs/2405.08366.

[18] Samuel Marks, Can Rager, Eric J. Michaud, Yonatan Belinkov, David Bau, and Aaron Mueller. Sparse feature circuits: Discovering and editing interpretable causal graphs in language models. *arXiv preprint arXiv:2403.19647*, 2024. URL https://arxiv.org/abs/2403.19647.

[19] Thomas McGrath, Matthew Rahtz, Janos Kramar, Vladimir Mikulik, and Shane Legg. The hydra effect: Emergent self-repair in language model computations. *arXiv preprint arXiv:2307.15771*, 2023. URL https://arxiv.org/abs/2307.15771.

[20] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, 2022. URL https://arxiv.org/abs/2202.05262.

[21] Aaron Mueller, Atticus Geiger, Sarah Wiegreffe, et al. Mib: A mechanistic interpretability benchmark. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 45069–45108, 2025. URL https://proceedings.mlr.press/v267/mueller25a.html.

[22] Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations*, 2023. URL https://arxiv.org/abs/2301.05217.

[23] Chris Olah, Arvind Satyanarayan, Ian Johnson, et al. The building blocks of interpretability. *Distill*, 2018. doi: 10.23915/distill.00010. URL https://distill.pub/2018/building-blocks/.

[24] Chris Olah, Nick Cammarata, Ludwig Schubert, et al. Zoom in: An introduction to circuits. *Distill*, 2020. URL https://distill.pub/2020/circuits/zoom-in/.

[25] Catherine Olsson, Nelson Elhage, Neel Nanda, et al. In-context learning and induction heads. *arXiv preprint arXiv:2209.11895*, 2022. URL https://arxiv.org/abs/2209.11895.

[26] OpenAI. Sharing ai progress in mathematics, October 2026. OpenAI research announcement.

[27] OpenAI. Mathematics manuscript collection, 2026.

[28] Soumadeep Saha, Akshay Chaturvedi, Saptarshi Saha, Utpal Garain, and Nicholas Asher. Kismath: Do llms have knowledge of implicit structures in mathematical reasoning? *Transactions of the Association for Computational Linguistics*, 14:1308–1328, 2026. doi: 10.1162/tacl.a.729. URL https://aclanthology.org/2026.tacl-1.59/.

[29] Adly Templeton, Tom Conerly, Jonathan Marcus, et al. Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet. *Transformer Circuits Thread*, 2024. URL https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html.

[30] Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. *arXiv preprint arXiv:2305.04388*, 2023. URL https://arxiv.org/abs/2305.04388.

[31] Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, et al. Causal mediation analysis for interpreting neural NLP: The case of gender bias. *arXiv preprint arXiv:2004.12265*, 2020. URL https://arxiv.org/abs/2004.12265.

[32] Kevin Wang, Alexandre Variengien, Arthur Conmy, et al. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. *arXiv preprint arXiv:2211.00593*, 2023. URL https://arxiv.org/abs/2211.00593.

[33] Jason Wei, Xuezhi Wang, Dale Schuurmans, et al. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, 2022. URL https://arxiv.org/abs/2201.11903.

[34] Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. *arXiv preprint arXiv:2309.16042*, 2023. URL https://arxiv.org/abs/2309.16042.